Biomedical Engineering Graduate Program

PhD Graduate Student Handbook

Academic Year 2025-2026

School of Biological and Health Systems Engineering

Arizona State University

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Introduction

Our graduate educational goal is to prepare graduates to work as biomedical engineers for the broad range of opportunities available in industrial, commercial, and academic organizations and to prepare graduates for continued learning experiences either in a formal graduate or professional program or in continuing education. Students will have core knowledge in, engage in independent research in, and disseminate knowledge in one or more of the following areas: biomaterials, biosensors, biomarkers and biomimetic materials; biomedical imaging; molecular, cellular and tissue engineering; neural and rehabilitation engineering; and/or synthetic and systems biology. Our alumni will be enabled to advance in their fields in academia or industry with their chosen careers as researchers, entrepreneurs, and engineers. We aim to provide exceptional graduate training in Biomedical Engineering that benefits our students, our community, our stakeholders, and society in line with the university's Charter. **We extend a warm welcome to all to the School of Biological and Health Systems Engineering (SBHSE) at Arizona State University!**

ASU Charter

ASU is a comprehensive public research university, measured not by whom it excludes, but by whom it includes and how they succeed; advancing research and discovery of public value; and assuming fundamental responsibility for the economic, social, cultural and overall health of the communities it serves.

Acknowledgement of Inclusion

The School of Biological and Health Systems Engineering (SBHSE) in the Ira A. Fulton Schools of Engineering (FSE) recognizes the intrinsic value of inclusion within academia. SBHSE commits to breaking down traditional structures to build an inclusive culture and community for our students, faculty, and staff to thrive within local and global biomedical engineering communities. SBHSE also recognizes diversity in many dimensions as a strength in innovation and developing the next generation of engineers. ASU's charter is an integral part of our standing as an institution and demonstrated by how we uphold, value, and cherish the diversity of our students and faculty members.

Academic Integrity

At Arizona State University academic honesty is expected of all students in all examinations, papers, academic transactions and records. The possible sanctions include, but are not limited to: appropriate grade penalties, loss of registration privileges, disqualification and dismissal. ASU strictly adheres to the academic integrity policy. This policy sets forth the ASU Student Academic Integrity Policy and appeal procedures. The policy can be found on the University Provost website. Additional information and resources can be found on the Ira A. Fulton Schools of Engineering website regarding Academic Standards and Academic Integrity. Students and faculty are also expected to adhere to the Arizona Board of Regents Code of Conduct.

Discrimination, Harassment, and Retaliation

ASU prohibits all forms of discrimination, harassment, and retaliation. To view ASU's policy please see https://www.asu.edu/aad/manuals/acd/acd401.html.

Title IX of the Education Amendments of 1972 is a federal law which provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy ACD 401 make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. For information on resources, visit the sexual violence awareness, prevention, and response website here.

Title IX protects individuals from discrimination based on sex in any educational program or activity operated by recipients of federal financial assistance. As required by Title IX, ASU does not discriminate on the basis of sex in the education programs or activities that we operate, including in admission and employment. Inquiries concerning the application of Title IX may be referred to the Title IX Coordinator or to the U.S. Department of Education, Assistant Secretary, or both. Contact titleixcoordinator@asu.edu or 480-965- 0696 for more information. Office located at 1120 S. Cady Mall, INTDSB 284. For information on making a report please go to www.asu.edu/reportit/.

Introduction

Objective of the Handbook

This handbook will provide you with the basic information needed throughout the course of study and assist you in navigating through the PhD program in SBHSE. The Handbook is the main source of information regarding policies, regulations, and academic requirements necessary to complete the PhD degree. We acknowledge that this handbook is not meant to be an exhaustive collection of all policies at ASU. Instead, students and faculty are encouraged to review the ASU Graduate Policies and Procedures regarding University policies on graduate programs. The relevant links are provided throughout this Handbook. You are responsible for being informed about all academic requirements of the graduate program. We also acknowledge that additional questions and concerns may arise that are not formally addressed in these sources. Our Advising staff, Graduate Program Chair, and Graduate Program Committee will be valuable assets as you progress through your degree. You are urged to maintain close contact with the Graduate Advisor and to seek additional information as the need arises.

Program Contacts

Graduate Program Chair: Sydney Schaefer, PhD
(Email: sydney<dot>schaefer<at>asu<dot>edu)

Assistant Director, Academic Services: Elizabeth Tripodi, M.Ed.
(Email: etripodi<at>asu<dot>edu)

Graduate Academic Success Advising Coordinator: Roberto Reynoso, M.B.A.
(Email: rjreynos<at>asu<dot>edu)

Program faculty: https://sbhse.engineering.asu.edu/faculty/

General Graduate Student Expectations

It is the responsibility of the graduate student to know and to observe all procedures and requirements as defined in this handbook, the Graduate Catalog, the Schedule of Classes, and the Guide to the Preparation of Doctoral Dissertation. A copy of the Schedule of Classes is available on-line at https://webapp4.asu.edu/catalog/. Graduate students are expected to be familiar with the Code of Conduct, which is available in the Office of Student Affairs. Violations of the Code of Conduct or incidents of dishonesty such as cheating in examinations, cheating in laboratory work or plagiarism is subject to university discipline whether committed by individuals or groups. **Graduate students are expected to demonstrate satisfactory progress** (see the definition of satisfactory progress in the Requirements for the Doctor of Philosophy Degree section below). They are also expected to maintain the highest degree of academic integrity, enthusiasm for their academic studies, and a high degree of professionalism. For more information about academic integrity, visit:
https://engineering.asu.edu/academic-integrity-for-students/

General Graduate Faculty Responsibilities

Faculty members serving as members of the SBHSE Graduate Faculty, especially those who are endorsed to the level of chair or co-chair, accept the responsibility of mentoring graduate students, and are expected to know and to observe the procedures and requirements defined in this handbook and the other publications listed above.

General Communication Expectations

It is the expectation of the student and all committee members, especially the Chair/Co-Chairs,to regularly engage in transparent communication about expectations, progress, and desired outcomes. All PhD students will have an advising hold each semester until they have successfully defended their prospectus where a check-in form needs to be completed to have the hold removed. This form is intended to facilitate critical conversations around a PhD student's progress and appropriate preparations for the next major milestone. It is expected that the Chair/Co-Chairs coordinate the efforts and expectations of the committee at large to ensure the PhD student(s) under the committee's mentorship can operate within clearly defined parameters and expectations.

General Safety

SBHSE is committed to providing a safe work environment for faculty, staff and students. Students are required to follow safe procedures in accomplishing their research and teaching assignments, in line with the Dean's Office of Infrastructure and Safety for the Ira A. Fulton Schools of Engineering. This Office is a department devoted to providing safety and services for all of the engineering schools, faculty, researchers, staff, and students. The primary goal of the FSE DO Infrastructure and Safety Team is to provide a central hub of resources and information to assist faculty, researchers, staff, and students in their day-to-day and long-term activities while ensuring a safe work and research environment. The Team is coordinated by the Dean's Office and is comprised of representatives in each of the engineering schools. Students are required to take a series of safety or safety refresher courses EVERY year.

College and University Procedures and Policies

All policies and procedures outlined in this handbook are in accordance with policy set by the Graduate College and Office of the University Provost.

Path to the PhD Degree

The student must accomplish several activities and milestones in the process of acquiring the PhD degree. The diagram below summarizes the chronological steps that must be followed in this process, with a suggested timeline. Students will consult with their faculty advisor directly for a more individualized timeline that is determined by a number of factors. The median time to completion of our program is 5 years for students who are enrolled full-time.

A “typical” timeline

Please note that the timeline varies between students, depending on their research, advisor, life circumstances, and many other reasons. The majority of our faculty-funded PhD students finish in 5 years or less.

Note: Students must graduate **within 10 years of admission**. However, an extension may be initiated in special circumstances with faculty advisor, department, and Graduate College approval via a petition. Please see the Graduate Academic Advisor for more information.

Goal of the Doctoral Program

The School of Biological and Health Systems Engineering (SBHSE) provides a comprehensive education to a multicultural student body covering many areas in biomedical engineering, preparing large numbers of engineers who will engage in industry, health-related and biomedical research, education, and entrepreneurship, thus addressing a wide selection of societal needs. These efforts support the university’s mission of advancing research and discovery of public value, and assuming fundamental responsibility for the economic, social, cultural and overall health of the communities it serves.

The Doctor of Philosophy degree is the highest university degree. It is granted to students upon evidence of excellence in research and the demonstration of independent, creative scholarship culminating in a dissertation. Coursework in the doctoral program focuses primarily on the engineering science concepts in the student's major and in certain basic sciences. The graduate research program introduces the student to the techniques, procedures and philosophical attitudes necessary for exploring unknown areas in his/her chosen profession. After receiving the degree, the student is able to identify areas within his/her major suitable for research; identify the current state of knowledge in these areas using literature search resources; propose plans for investigating the area; apply fundamental principles of science and engineering to complete the investigation, and teach these skills to others who follow. The student is taught the scientific method by intensely studying a specific research topic. This also yields a more in-depth knowledge of his/her professional major. Often included in the graduate educational experience is an opportunity to teach undergraduates by preparing selected lectures in undergraduate courses, assisting in undergraduate laboratories or serving as tutors or mentors.

We also acknowledge that selecting a PhD program involves a lot of decisions and factors. In addition to our cutting-edge research and world-class facilities, which you can learn about on our [website](), we want you to know that:

100% of our PhD graduates reported **receiving 1 or 2 job offers after graduation***.

*based on responses from 2023-24 ASU First Destination Survey

Prospective Students

Of paramount importance to a successful doctoral program is the selection of a suitable research topic and faculty advisor. Prospective students are encouraged to visit the SBHSE website (https://sbhse.engineering.asu.edu/) to learn more about our research areas and our faculty expertise. Prospective students who are interested in pursuing their PhD in BME at Arizona State University should reach out directly to faculty of interest and inquire about available funding opportunities and to learn more about their ongoing research. Since most PhD students are financially supported by their faculty advisors (see “General Admissions Requirements - Other information”), there is not always a perfect match between the student and faculty member in terms of research interest. Thus, early communication between a prospective student and a faculty member prior to (or concurrent with) applying to the PhD program is encouraged.

In this PhD program, original work is required for the Doctor of Philosophy degree. One or more peer-reviewed research publications or presentations should result from the research project. Throughout their program of study, the student is encouraged to actively participate in efforts to acquire funding in support of the advisor's research program. The student should assist the research advisor in the preparation of grant proposals to local, state and national agencies seeking funding for the project, and should also consider applying for their own funding whenever appropriate.

The student-advisor relationship is a vital one during the PhD years, and often continues well beyond them. Each such relationship is unique, and usually offers personal and professional benefits beyond the conduct of the PhD research. These benefits might include meeting important post-degree job contacts, advice on professional development and training in non research-related professional skills (e.g. teaching). It is expected that in most circumstances student-advisor disagreements will be minor and will be amicably resolved by those involved. In the uncommon instances that attempts to resolve disagreements are unsuccessful, the student and advisor are encouraged to meet with the graduate student academic advisor and/or the program chair for further assistance in resolving any difficulties. Dissertation committee members may also serve as informal mentors throughout the course of the degree.

As such, we encourage students, once admitted to our program, to check in with their committee members more than once a semester to update them on their progress, seek advice, etc.

General Admission Requirements

Regular Admission

To be eligible for regular admission, the student must have a Bachelor's degree in Bioengineering, Biomedical Engineering, or equivalent coursework (see below). Applicants normally will have a minimum grade point average (GPA) of 3.0 out of a total possible 4.0 or equivalent in the last 60 hours of their first bachelor's program. Students entering with a master's degree are required to have a minimum GPA in their master's degree coursework of 3.0 out of a possible 4.0. Foreign students must also submit test scores from the Test of English as a Foreign Language Exam (TOEFL), International English Language Testing System (IELTS), or Duolingo in compliance with the current requirements of the Graduate College. TOEFL scores should be close to 100 for admission.

Submission of GRE scores is optional for Biomedical Engineering PhD program applications. Applications are evaluated using a holistic review process that considers the multiple, intersecting factors – academic, nonacademic, and contextual – that uniquely define each applicant. This process can include, but does not require, consideration of GRE scores. Thus, an absence of GRE scores will not be viewed negatively during the application review process. Applicants who chose to have their scores considered as a supplement to their application should submit them to ASU's Graduate Admission Services and indicate in their personal statement how the scores supplement their application.

Regarding letters of recommendation, applicants are encouraged to seek doctoral-level references from their own academic and/or professional institutions who can speak to their academic potential.

Regular Admission with Unmet Prerequisites

Regular admission may also be given to students with a Bachelor of Science and/or Master's degree in another discipline. In this case, however, the student may be required to take specific courses to fulfill the needed pre-requisite coursework. These courses are often in addition to the graduate program of study. The letter of admission specifies the necessary courses that must be completed before the student is awarded the graduate degree based on unmet prerequisites. Students will be required to complete any unmet prerequisites within the first year of their admission to the Biomedical Engineering PhD program. .

Students without a Bachelor's or Master's degree in Biomedical Engineering and without the equivalent of the following courses in their undergraduate program of study may have unmet prerequisites and may require some additional core competencies needed for graduate study in Biomedical Engineering. These courses must be completed in addition to the required graduate coursework. In addition, the student's supervisory committee may outline additional transition program requirements to ensure that the student can successfully pass the Comprehensive Examination.

Pre-requisite Coursework

Mathematics and Basic Sciences

Mathematics: Calculus through Differential Equations (e.g. MAT 270, 271, 272 AND 275; typically at least 12 semester hours credit total).

Physics: One year of calculus-based physics including laboratory (typically 8 semester hours).

Biology: Minimum of one "General Biology" course (typically 4 semester hours).

Physiology: Minimum of one "Physiology" course (typically 4 semester hours).

Chemistry: Minimum of one Chemistry course including laboratory (typically 4 semester hours).

General Engineering Fundamentals

Students without the equivalent courses must complete additional course work in **four** of the following six topics:

Thermodynamics
Fluid Mechanics
Mechanics of Rigid Bodies
Electrical Networks
Signals and Systems
Materials Science and Engineering

As mentioned above, additional coursework beyond what is listed here may be assigned to best support the student and faculty member's area of research. Any additional coursework will become part of the student's degree requirements and must be completed as part of the graduate program of study**.**

Students and their faculty advisor(s) may work with the Graduate Academic Advisor to resolve potential unmet prerequisite coursework based on their transcripts, depending on the nature of prior coursework and the anticipated research. The exact courses chosen for remediation will vary depending on the specific field of research. The program makes every attempt to ensure coursework being assigned is relevant to a student's area of study and will ultimately benefit their degree progress.

Provisional Admission

Applicants with scholastic records below the standards for regular admission may be admitted provisionally in certain special cases at the discretion of the departmental graduate committee with the approval of the chair of the graduate committee and the department chair. A student admitted with provisional status must follow the provisional terms as outlined in the admission letter, typically earning a 3.25 out of a 4.0 in the first semester in nine (9) graduate level credits.

Full-time provisional students must take a minimum of nine (9) hours during their first semester in residence. Part-time provisional students may take fewer than nine (9) hours of coursework during their first semester. Failure to do this will result in suspension from the program. Students who meet this requirement are reclassified as a regular graduate student and the regulations governing academic performance for regular students apply. **It is the student's responsibility to see that their status is changed from provisional to regular after having successfully completed these requirements.** Please contact your Graduate Advisor when you have fulfilled the provisional requirements.

Non-Degree Seeking Pathway

It is not uncommon for prospective students interested in our PhD program to have unmet prerequisites in their prior coursework, especially if the student comes from another discipline. In some cases, and for other possible reasons, students may wish to enroll at ASU as a non-degree seeking graduate student. This option gives students the ability to take ASU classes at either the undergraduate or graduate level (may need specific department approval) to work on pre-requisites, demonstrate aptitude for the degree program, and begin networking with faculty. Up to 12 credits of graduate-level coursework taken as a non-degree seeking student can be transferred to the PhD program provided the credits meet program requirements. Please note that the non-degree seeking option is not necessarily available to all students. For example, international students attending a university on an F-1 or J-1 visa status are typically not able to enroll as a non-degree student due to visa restrictions that mandate enrollment in a degree bearing program. It is the student’s responsibility to review and be aware of all requirements associated with the non-degree seeking pathway. There is no specific advising support associated with the non-degree enrollment option, but the SBHSE Advising Office is happy to offer advice and guidance on how course choices align with PhD degree requirements and provide information on admission requirements.

Please note that the non-degree seeking pathway does not provide a guarantee of admission. Rather, students are encouraged to consider it as one possible opportunity to gain program specific knowledge and network to motivate a hiring offer (and thereby admission offer) from a faculty member.

Other Admissions Information

Eligible applicants are admitted into the PhD program under the supervision and financial support of a faculty advisor with a primary appointment in the School of Biological and Health Systems Engineering (SBHSE) and with membership to its graduate faculty. Eligible applicants can be admitted under the supervision and financial support of a faculty member with a primary appointment outside of SBHSE, either in another academic unit within ASU or at an established clinical partner, who agrees in writing to the processes and procedures outlined in this Handbook. In all cases, financial support is provided through the faculty advisor in the form of a Graduate Research Assistantship (GRA) during the academic year at minimum, with the advisor's funding confirmed by their research administration official prior to admission. The only acceptable forms of funding for student stipends, tuition, and benefits are: 1. financial support from the faculty advisor; 2. external academic fellowships (e.g., NSF GRFP, GEM Fellowship); and 3. ASU-funded fellowships. In the case of fellowships, the faculty advisor commits to supplementing the fellowship if needed to meet SBHSE minimum stipend level, tuition, and benefits, in addition to funding any years remaining after the completion or loss of the fellowship. Unacceptable forms of funding include self-funding through an industry salary or personal/family finances (e.g., quarters from Aunty).

Tuition, Fees, and Residency

Tuition is set by ASU and the Arizona Board of Regents each year. View the general Tuition and Fees Schedule, or calculate a more specific estimate of charges using the ASU Tuition Estimator. Information on residency requirements can be found at Residency for Tuition Purposes.

All amounts shown in the Tuition and Fees Schedules or in other University publications or web pages represent tuition and fees as currently approved. However, Arizona State University reserves the right to increase or modify tuition and fees without prior notice, upon approval by the Arizona Board of Regents or as otherwise consistent with Board policy and to make such modifications applicable to students enrolled at ASU at that time as well as to incoming students. In addition, all tuition amounts and fees are subject to change at any time for correction of errors. Finally, please note that fee amounts billed for any period may be adjusted at a future date.

For details about current and past Graduate Program Fees, click here. As noted below in Financial Support, fees are typically covered by the student themselves.

Requirements for the Doctor of Philosophy Degree

The Graduate College sets certain general requirements for the Doctor of Philosophy degree. In addition to these general requirements, the department sets specific program requirements, which exceed those imposed by the Graduate College. This section outlines both the general requirements specified by the Graduate College and the additional requirements specified by the Biomedical Engineering Program.

Grading

Grades are assigned in graduate courses in compliance with the current definitions as set by the university. For more information, please see: https://students.asu.edu/grades

Repeating ASU Courses

Graduate students (degree or nondegree) may retake any course at any level at ASU, but all grades remain on the student transcript as well as in GPA calculations.

Good Standing and Satisfactory Academic Progress

A student who has been admitted to a graduate degree program in Engineering, with either regular or provisional admission status, must maintain a 3.0 or higher grade point average (GPA):

1. in all work taken for graduate credit (courses numbered 500 or higher),
2. in the coursework in the student's approved program of study, and
3. in all coursework taken at ASU (overall GPA) post baccalaureate.

A student will be placed on academic probation if one or more of the student's GPAs listed above falls below 3.0. Students will be notified by email when placed on academic probation.

A student will earn academic good standing by obtaining a 3.0 or better in the GPAs listed above by the time the next nine hours are completed. Coursework such as research and dissertation registration that are for Z or Y grade cannot be included in these nine hours.

A student may be recommended for dismissal from a graduate program if the student fails to increase all of the GPAs listed above to 3.0 or better by the time he/she completes at least nine credit hours.

For PhD students, adhering to the appropriate timing of key degree milestones is critical to student success and timely degree completion. Thus, the definition of satisfactory academic progress includes completion of the initial attempt at the comprehensive exam by the end of the first year, completion of the practicum requirement during the second year, and having a working draft of the prospectus by the end of year two in anticipation of the prospectus defense in the third year. Satisfactory academic progress is further defined as making a good faith effort to fulfill the expectations of their personalized learning plan, finalizing a committee in a timely manner, and submitting the iPOS prior to 50% completion of the degree. Students who fail to meet the expectations of satisfactory academic progress may be recommended for dismissal from the graduate program. Additional requirements may be imposed by the supervisory committee.

A student may appeal actions concerning dismissal by petitioning the departmental unit in which they are enrolled.

Misconduct

The highest standards of academic integrity are expected of all students. The failure of any student to meet these standards may result in suspension or expulsion from the university and/or other sanctions as specified in the academic integrity policies of the individual colleges. Violations of academic integrity include, but are not limited to, cheating, fabrication, tampering, plagiarism, falsification or misrepresentation of data or facilitating such activities. The university and Fulton Schools of Engineering academic integrity policies are available online.

Graduate Credit Courses

Courses at the 500, 600 and 700 levels are graduate credit courses. Courses at the 400 level satisfy graduate degree requirements when appearing on an approved plan of study. There is a limit of 6 credits of 400 level courses that can be included on the plan of study.

Transfer Credit

Transfer of credit is the acceptance of credit from another institution for inclusion in a program of study leading to a degree awarded by ASU. Transfer of credit can also apply to credits taken at ASU as a non-degree student.

Transfer credits may not be applied toward the minimum degree requirements for an ASU degree if they have been counted toward the minimum requirements for a previously-awarded degree.

The number of hours transferred from other institutions may not exceed 12 hours and must be at 500 level with a B or better and within 3 years of admission. Up to 12 semester hours of credit taken at another institution must not have been counted toward a previous degree and may be counted toward the minimum semester hours required for a specific ASU doctoral degree program. In all cases, the inclusion of transfer courses on a program of study is subject to approval by the academic unit and the Graduate College.

Students with a Master's Degree in Biomedical Engineering from another institution may transfer up to 30 semester hours of credit towards the course requirements for the Doctor of Philosophy degree with the approval of the departmental graduate committee and Graduate College. The overall course credits, however, must conform to the above requirements. The student must have credit for the Biomedical Engineering core courses, and sufficient Biomedical Engineering graduate electives regardless of the institution where they were taken. The Graduate College requires that at least 54 hours of the program of study are taken at ASU. The determination of applicable courses will be made by the Chair of the committee and should be completed within the first semester of coursework.

Certain types of graduate credits cannot be transferred to ASU, including the following:

1. credits awarded by postsecondary institutions in the United States that lack candidate status or accreditation by a regional accrediting association;

2. credits awarded by postsecondary institutions for life experience;
3. credits awarded by postsecondary institutions for courses taken at noncollegiate institutions (e.g., government agencies, corporations, and industrial firms);
4. credits awarded by postsecondary institutions for noncredit courses, workshops, and seminars offered by other postsecondary institutions as part of continuing education programs;
5. credits given for extension courses; and
6. credits completed before the posting of a bachelor's degree.
7. Acceptable academic credits earned at other institutions that are based on a unit of credit different from the ones prescribed by the Arizona Board of Regents are subject to conversion before being transferred to ASU.

Transfer credits must be acceptable toward graduate degrees at the institution where the courses were completed. Only resident graduate courses (at the institution where the courses were completed) with an "A" (4.00) or "B" (3.00) grade may be transferred. A course with the grade of pass, credit, or satisfactory may not be transferred. Additionally, transfer credits must be within the six-year time limit to be used on a plan of study.

Official transcripts of any transfer credit to be used on a plan of study must be sent directly to the Graduate Admissions Office from the Office of the Registrar at the institution where the credit was earned.

Course Load

Course load is not to exceed 13 semester hours (without approval) of credit during each of the two semesters, 6 semester hours during each 6-week summer session or 9 semester hours of credit during the 8-week summer session or a petition must be submitted. All students must register for a minimum of one credit each semester to continue as a graduate student at ASU.

All graduate research assistants and associates (RA/TA's) must enroll for 12 credit hours (this may include research credit hours) during each semester of their employment. This departmental requirement generally exceeds the Graduate College minimum. The hours cannot include audit enrollment. A half-time (50%) graduate assistant or associate working 20 clock hours per week may not register for more than 12 hours of coursework each semester; a one-third time (33%) assistant or associate for more than 13 hours and one-quarter time (25) assistant or associate for more than 15 hours. A graduate assistant or associate (RA/TA) may petition to take more than 13 credit hours for a semester.

During the summer session, graduate assistants and associates must be enrolled in at least one credit in coursework related to their degree program in each of the summer session terms. During the summer sessions, graduate assistants employed 25% time may enroll for a maximum of 6 semester hours during a 6-week session or 9 hours during an 8-week session; those employed 50% may enroll for a maximum of 5 hours during a 6-week session or 7 hours during the 8 week session and those employed 100% time may enroll for a maximum of 3 hours during the 6 week session o 4 hours during the 8-week session.

All graduate students doing research, working on thesis or dissertations, taking comprehensive examinations, or using university facilities or faculty time, must be registered for a minimum of one hour of credit that appears on the program of study or is an appropriate graduate level course. For additional information, visit the Graduate College handbook for RA/TA's at

https://graduate.asu.edu/ta-ra

Dissertation Supervisory Committee

As described in the Graduate Catalog, each student interacts with one committee appointed by the dean of the Graduate College: the Dissertation Supervisory Committee. This committee will oversee the student's curriculum and research progress as well as oversee the Prospectus and Dissertation Defense requirements. The Dissertation Supervisory Committee should be formed by the end of the third semester. **The form to complete one's committee can be found here.**

Committee chair

The committee chair is generally the research advisor (*aka* faculty advisor), unless the research advisor is not endorsed to the level of Chair by the Graduate College; if this is the case, the research advisor should serve as co-chair and a co-chair with relevant research expertise will be selected (who has a primary appointment in SBHSE). **The committee chair (or co-chair, should the primary faculty advisor be outside of SBHSE) must be a tenured or tenure-track faculty member in SBHSE or a member of the Graduate Faculty in Biomedical Engineering (BME) with endorsement to co-chair or chair. When the primary faculty advisor is endorsed to the level of co-chair (i.e., a faculty member outside of SBHSE), a 2nd co-chair from among the tenured/tenure track SBHSE faculty must be appointed, preferably one with knowledge of the student's research area.** Non-BME faculty *may* receive endorsement to chair as decided by the Graduate Program Committee based on a number of criteria, including but not limited to prior history of BME PhD student mentorship to completion (i.e., graduation), student success (i.e., the timeline to completion), and other programmatic involvement by the faculty member in question.

As noted above, because the committee chair is generally the faculty advisor, it is most common that the individual(s) serving as committee chair and/or co-chair is appointed at the recruitment stage prior to a student enrolling in the program. Individuals interested in recruiting prospective PhD students from the Biomedical Engineering degree program should complete the Application to join SBHSE PhD Graduate Faculty to initiate the steps for recruiting a particular student.

Committee members

The committee should be comprised of at least four (but up to six, in some cases) members; one of these members will serve as the chair (or two as co-chairs), along with other faculty or experts in the student's field of research. Fifty percent (50%) of the committee must be tenured/tenure-track SBHSE faculty. The members of the dissertation committee should have the necessary knowledge and skills to advise the student during the formulation of the research topic and to provide input throughout the completion of the research and the dissertation. The committee must be approved by the dean of the Graduate College before the student may schedule their prospectus defense. If there needs to be any additions to the membership of the committee after the committee has been appointed, the student must re-submit the supervisory committee form with the name and signature for the new addition and receive the approval through the same process as initial submission. If a committee member needs to be removed, the student will need to delete the committee member from their Interactive Plan of Study (iPOS). It is the student's responsibility to ensure they have communicated to the removed member that they no longer need said member to serve in that committee role.

Committee member attendance at defenses

While it is desirable that all members of a student's supervisory committee be physically present with the student at the final oral dissertation defense, there are situations (e.g., faculty travel, faculty emergencies and/or faculty leave) that may necessitate holding a defense with one or more committee member(s) absent. The Graduate College office has established the following policies and procedures for such cases, as described here:

1. A minimum of 50% of the student's official committee must be physically present with the student at the defense. If at least 50% of the committee cannot be physically present, the defense must be rescheduled.

2. The chair or one co-chair must be physically present at the defense. If this is not possible, the defense must be rescheduled. The student cannot submit a committee change after the defense is scheduled.

3. A committee co-chair or member who cannot be physically present at the defense may participate in the defense in one of three ways. These options are listed in the order of preference:

   a. The absent committee member videoconferences into the defense location.*

   b. The absent committee member teleconferences into the defense location.*

   c. The absent committee member provides a substitute to be physically present (approved by the committee chair, the head of the academic unit & the Graduate College) for the defense only. The substitute must be someone who is approved to serve on graduate supervisory committees for that program. The absent committee member should provide the substitute questions, in writing, to be asked at the defense. The substitute, although respecting the opinions expressed by the regular committee, must be free to use his/her judgment in voting on whether the student passes or fails the defense.

*The defense location must have the necessary equipment to accommodate video/teleconference materials.

*Students must provide a copy of their document and any other supporting presentation materials to the substitute committee member at least 5 working days in advance of the defense.

Absent Committee Member Signature Instructions: If a committee member will be absent from the defense, the academic advisor or committee chair/co-chair must notify the Graduate College as quickly as possible and before the defense takes place. In order to assign a substitute, please be prepared to provide the Graduate College with the full name and email address of the faculty member who will serve as the substitute.

Please contact the Graduate College office at (480) 965-3521 or send an email to Grad-GPS@asu.edu if you have questions or concerns regarding these procedures.

Interactive Plan of Study

The student is required to file an interactive plan of study (iPOS) with SBHSE and the Graduate College no later than after 50% completion of their degree plan credit hours. It is highly recommended to file the iPOS after the first semester. The iPOS will be available on MyASU. Changes in the planned program may be made with the approval. A step-by-step guide from the Graduate College for creating your iPOS is available here.

Sample Plan of Study

The sample plan of study included in this handbook is intended for **informational purposes only.** It serves as a general guide to help students understand the typical course sequencing and timeline within the program. Each student's academic journey is unique, and individual plans may vary based on course availability, prior coursework, research interests, and academic progress. Students should consult regularly with academic advising to develop and maintain an individualized plan of study that aligns with their goals, circumstances, and curricular requirements.

Term	Requirement	Credits
Year 1, Fall (semester 1)	BME 5XX course	3
	Quantitative course	3
	Life Sciences course	3
	BME 591	1
	BME 792 (Research)	2
Milestone:	Submit initial iPOS with courses	By end of first semester
Year 1, Spring (semester 2)	BME 5XX course	3
	Quantitative course	3
	Life Sciences course	3

Term	Requirement	Credits
	BME 591	1
	BME 792	2
Milestone:	Complete Comprehensive Exam	End of semester 2
Year 2, Fall (semester 3)	Technical elective	3
	BME 780 (Practicum)	3
	BME 591	1
	BME 792	5
Year 2, Spring (semester 4)	Technical elective	3
	Technical elective	3
	BME 591	1
	BME 792	5
Milestone:	Complete teaching practicum	By end of year 2
Year 3, Fall (semester 5)	BME 591	1
	BME 792	11
Year 3, Spring (semester 6)	BME 591	1
	BME 792	11
Milestone:	Prospectus defense	By end of year 3
Year 4, Fall (semester 7)	BME 792	12
Year 4, Spring (semester 8)	BME 799 (Dissertation)	12
Milestone:	Dissertation defense	By Graduate College deadline

This example plan of study is for a student who does not have a conferred master’s degree or any applicable transfer credit. This plan of study also does not include deficiency requirements that may need to take precedence in the first year of study. It is the student’s responsibility to ensure their intended plan of study factors in any unique components particular to their situation.

Course Requirements

Total Requirements

A minimum of eighty-four (84) semester hours is the total course/seminar/research/ dissertation requirement. Total hours are 84 semester hours of graduate- level courses.

1. Doctoral students are required to complete 6 credits from each of the following 3 areas (at least 18 credits):

   BME Courses (BME Prefix) – 6 credits

   Life Science/Biology Courses – 6 credits

   Quantitative Math or Engineering – 6 credits (from the approved list found here)
2. Technical Electives - 9 credits (courses relevant to research focus)
3. BME 591: Seminar - 5 credits
4. BME 780: Teaching Practicum - 3 credits
5. BME 799: Dissertation - 12 credits (override required; student is eligible to enroll after successfully passing the prospectus defense.)
6. BME 792: Research - 37 credits (minimum)

** A maximum of 6 of the above credits may be at the 400-level

Teaching Practicum

The teaching practicum is a mentored teaching experience that the student participates in in cooperation with a faculty member. The purpose of this practicum is to provide students with training and experience in instructional design within a Biomedical Engineering curriculum, consistent with expectations of student teaching at peer PhD programs in Biomedical Engineering at other universities. At a minimum, the student will experience several elements of teaching: preparation and delivery of at least 3 class sections, holding office hours, selection/creation and evaluation of student work, design, preparation and evaluation of one examination, and exposure to inclusive pedagogical teaching principles. In addition, the student and the course instructor will come up with a specific plan for providing the student with feedback regarding his or her performance during the experience. Once the experience is completed, the student and the instructor will file brief reports with the graduate chair describing the practicum.

Timing

To ensure timely degree progression and to facilitate protected time for dissertation research, the teaching practicum should be completed in the second year of a student's course of study in the PhD program. Failure to complete the practicum in Year 2 without prior authorization from the Graduate Program Committee will result in a failure to maintain satisfactory academic progress, and the student will be placed on a probationary status.

Course selection process

PhD students will complete their teaching practicum in one of our advanced biomedical fundamentals courses, which are:

- BME 510: Biomechanics/Human Physical Capability
- BME 598: Advanced Modeling of Transport Phenomena for BME
- BME 598: Signals and Systems Mastery: Practical Applications in BME
- BME 598: Biomolecular and Genetic Engineering
- BME 598: Applied Programming: Data Modeling and Analysis
- BME 598: Biomaterials Design and Application

The exact course assigned to the practicum student will be made in consultation with the Associate Director of Academic Excellence, the Graduate Program Committee, and the student's Faculty Advisor. Course selection may be informed by the goals of the student's personalized learning plan, along with course support needs and related factors.

Hosting the practicum requirement in the program's "core" courses helps ensure a standardized and structured experience centered on upskilling a student's knowledge of instructional design principles, course delivery methods, and content knowledge of a fundamental BME topic.

Registration

In order to register for practicum (BME 780, 3 credits), the student will first have to file a Teaching Practicum Course Definition form with the advising office. This ensures that the student and faculty member have reached agreement over expectations for the practicum experience. The form is available here. Students completing their Teaching Practicum will be enrolled in a corresponding Canvas course that semester for the submission of their deliverables. Students will also be expected to complete training similar to IA/TAs at the beginning of the semester in which they complete the practicum requirement.

Foreign Language Requirement

None.

Comprehensive Examination

Purpose and Committee Composition

The comprehensive examination is designed to test the extent to which a student demonstrates knowledge and understanding in BME coursework and corresponding prerequisites as they pertain to the student's specific research area, and demonstrate sufficient foundational knowledge to develop a research plan in their selected sub-discipline of Biomedical Engineering as the next step. The Comprehensive Exam is separate from the Prospectus, and should *not* focus on or evaluate the student's proposed dissertation research. Students are evaluated by a

Comprehensive Exam Committee composed of a diverse group of SBHSE tenure/tenure-track faculty with expertise across all research thrusts.

Exam Timeline and Preparation

The student is expected to take the comprehensive examination at the conclusion of their first year in the PhD program. The comprehensive exam will be administered on a set date established by the Comprehensive Exam Committee for all first-year PhD students and is administered in two parts: a written portion and an oral portion. Questions will be prepared in advance by the Comprehensive Exam Committee and cover topics that relate to SBHSE's research thrusts and core competency expectations. Students will be required to select a set number of questions from these provided options at the beginning of the exam. This allows students to select questions within their area of proficiency and research focus, but also ensures all PhD candidates are demonstrating deep understanding of fundamental biomedical engineering concepts.

The duration of the exam, along with the final list of materials allowable for use during the comprehensive exam, will be determined collectively by the Comprehensive Exam Committee. If a student has accommodations on file with the university's office of Student Accessibility and Inclusive Learning Services regarding exam time limits or materials, those instances will be handled on a case-by-case basis. Upon completion of the written exam time period, the oral exam portion will begin where the Comprehensive Exam Committee will review the written responses and 'cross-exam' each student individually for a set amount of time.

At the conclusion of the exam, a student's written and oral performance will be evaluated by the Comprehensive Exam Committee using an established rubric. The Comprehensive Exam Committee chair will compile all exam materials and evaluations to then pass on to the Graduate Program Committee, along with an outcome recommendation from the Comprehensive Exam Committee: pass, retake, or fail.

Once all materials, evaluations, and recommendations have been received by the Graduate Program Committee, a second round of review will take place. The Graduate Program Committee will meet with each student's Faculty Advisor to discuss the results of their student's comprehensive examination along with the recommendations of the review committees. The Graduate Program Committee and the student's Faculty Advisor will collectively determine the outcome of the exam and develop an appropriate personalized learning plan to further develop a student's skillset, serve as a roadmap to prepare for the Prospectus, and address any deficiencies identified in the comprehensive examination process.

Potential Exam Outcomes and Next Steps

Pass: If a decision of Pass is rendered, the Graduate Program Committee and Faculty Advisor will develop a personalized learning plan to guide the student on expectations for the following year (Year 2) to help the student prepare for the next milestones (Teaching Practicum and Dissertation Prospectus). Students who pass the comprehensive exam at the end of Year 1 are cleared to move forward with their teaching practicum and prospectus work in Year 2.

Retake: If a decision of Retake is rendered, the Graduate Program Committee and Faculty Advisor will develop a personalized learning plan that clearly articulates the deficiencies or concerns along with a remediation plan for how the student will address those concerns over the course of Year 2. At the end of the second year, the student will be permitted to retake the comprehensive exam on the set date following the same format as described above. There are only two possible outcomes after a student's second attempt: Pass or Fail. If a student passes the second attempt at the exam, they will proceed forward with the guidance outlined in the **Pass** section above. If the student does not pass the second attempt, they will proceed forward with the guidance outlined in the **Fail** section below.

Fail: A decision of Fail can only be rendered upon a student's second attempt at the comprehensive exam. In the cases where the student is on their second attempt of the comprehensive exam and the decision of Fail is rendered, this decision is final and the student will be dismissed from the program. No appeals will be granted following a failing result on a student's second attempt. If a student is dismissed from the program following two unsuccessful attempts at the comprehensive exam, they may be eligible to apply for a **Masters in Passing**, pending eligibility.

The Comprehensive Exam Committee, together with the Graduate Program Committee and the Faculty Advisor, reserves the right to render a decision of Fail to a first year student upon their first attempt at the exam should the student's performance be well below expectations and unable to be remediated even through a personalized learning plan. In the rare cases where this might occur, the student may petition the Graduate Program Committee to be permitted to re-take the exam, providing justification to be allowed a second attempt.

Dissertation Prospectus

Guidelines and Tips for the Dissertation Prospectus

The final outcome and intent of the prospectus should be viewed and interpreted as a contract between the PhD student and their dissertation committee on what is expected in their final defense of their dissertation. Thus, regardless of the results of their subsequent research (e.g., null findings), the student's level of work should be considered acceptable so long as they completed what was agreed upon by them and their committee when they successfully pass their prospectus.

A written component (rough draft of a research proposal) should be reviewed by the student's research advisor for approval of content prior to scheduling the oral presentation. The oral presentation of the dissertation prospectus is made to the student's dissertation committee.

The student will work with their advisor(s) to generate specific aims and write their dissertation proposal around it, in the **format** of a NIH R21 Research Strategy (6 pages) plus a one-page Specific Aims. Examples of this format can be found here. References are not included in the page limit, and should be provided in a separate "References" section in the overall prospectus document. Though not required, students are encouraged to informally meet with their committee members as well to discuss their research ideas, preferably several times during the time period between the Comprehensive Exam and the Prospectus. The **content** of the written + oral prospectus should cover what the student plans to complete for their dissertation, as well as a detailed plan of how they will achieve this plan.

It is the responsibility of the student to communicate with committee members directly to select the date/time of the prospectus. Students are encouraged to use scheduling tools like When2Meet or Doodle, and to plan well in advance (2-4 months out) to accommodate committee members' schedules. Prospectus exams are closed to the public, with only committee members present.

Once a date and time is selected in consultation with the committee members, the student requests a room for the selected date and time with the Prospectus Scheduling form. (Students are encouraged to email sbhse@asu.edu for further assistance if they do not get a response within 48 hours of completing the form, excluding weekends).

Students must also prepare a presentation of their prospectus, to be presented during the prospectus exam itself. The student's presentation should take advantage of appropriate audio-visual aids and should be limited to no more than 50 minutes. Copies of the written dissertation prospectus must be distributed to all members of the student's dissertation committee **at least one week** prior to the oral presentation.

In the oral examination, the student is expected to defend their prospectus and justify that the proposed research is of the acceptable quality and magnitude consistent with quality doctoral education. Following the oral presentation of the research proposition, members of the student's dissertation committee will remain to ask questions of the candidate regarding his/her/their proposed research.

Generally, the oral discussion of the dissertation prospectus is limited to three hours. If necessary, however, the proceedings may be adjourned and rescheduled for a mutually convenient date within one week. Only one adjournment is permissible.

After questioning, the candidate is excused from the room while the dissertation committee conducts its deliberations. The decision regarding whether or not the dissertation prospectus is acceptable is the decision of the dissertation committee alone. The student's dissertation committee conveys its evaluation of the acceptability of the dissertation prospectus to the chair of the departmental graduate committee by completing the Report of Dissertation Proposal/Prospectus form, which can be obtained through the SBHSE Advising Office. The outcome of the prospectus is based on majority vote in line with the Faculty Advisor vote. Committee members may voice disagreement and vote accordingly on the form, however a uniform overall decision must be reached at the conclusion of the deliberations. If a committee is split and/or encountering difficulties reaching a unified "pass" or "fail" decision, it is recommended for the "revision" option to be used in these cases with clearly communicated expectations from the committee as to the work that needs to be refined before scheduling a second defense. A recommended rubric for evaluating the prospectus can be found here.

If the student's dissertation prospectus is considered acceptable as a "pass", the chair of the departmental graduate committee will recommend to the Graduate College that the student be advanced to PhD candidacy status. At this stage the student may also apply for a Master's in Passing.

In the rare instance where the supervisory committee renders a decision of "fail", the supervisory committee must provide written detail justifying the decision. A decision of "fail" may result in dismissal of the student from the doctoral program, upon review of the Graduate Program Committee. If a student is dismissed from the program, they may be eligible to apply for a Masters in Passing.

Note: Students have a maximum of two (2) attempts to complete the prospectus defense successfully, upon approval of their committee and head of the graduate program. A second attempt should occur no sooner than one month following the first defense and no later than one calendar year from the date of the first defense. In the event that a second defense becomes necessary, the supervisory committee is responsible for establishing and communicating clear expectations and guidance in a documented prospectus revision plan provided to the student, the SBHSE advising office, and head of the graduate program. This revision plan must include an agreed upon re-examination timeframe factoring in the scope of expected revisions (e.g. the revisions should take no more than four months, so the deadline for the second defense should be set four months following the first defense). This plan will need to be submitted to the SBHSE Advising Office no later than **one week** (7 days) following the first prospectus defense date.

Master's in Passing for Direct-to-PhD Students

Direct-to-PhD students (i.e., those without a prior MS degree in BME) can apply for a master's in passing (MIP). The master's degree in passing will be the MS degree. The degree must be requested by the student through their Graduate Academic Advisor. The student must have completed 30 credits (following the BME MS degree requirements) with at least a 3.0 GPA and have passed at least the written portion of the PhD Comprehensive Exam per the Graduate College requirements. After completing 30 credits and their PhD Comprehensive Exam (form and report are required to be submitted to the academic advisor), students can work with their academic advisor to have the proper form submitted to Graduate College. Once that is approved, the student will need to complete the MS iPOS that lists the 30 credits completed and apply for graduation.

Once the MIP is awarded, students are still active in the PhD program unless they withdraw or are dismissed from the program.

Admission to Candidacy

PhD students achieve candidacy status in a letter from the dean of the Graduate College once they have:

1. Passed the comprehensive examinations; and
2. Successfully defended the dissertation prospectus.

Completion of the Written Dissertation

The written dissertation forms the culminating experience of the doctoral program. This document should reflect substantial effort on the part of the candidate to enhance the boundaries of knowledge in a field of relevance to biomedical engineering. The work should be novel and original, and of sufficient quality to merit publication in a peer-reviewed journal in the candidate's chosen area of research. In addition to carrying out the research and writing it up, the BME Graduate Program and the Graduate College have several requirements that must be met.

The written doctoral dissertation is based on an original and substantial scholarly work that constitutes a significant contribution to knowledge in the students' discipline. The dissertation research must be conducted during the time of the students' doctoral studies at ASU, under guidance of ASU Graduate Faculty and in accordance with the Graduate College policies and procedures. The composition of the dissertation is defined by the degree program with the approval of the Dean of the Graduate College. In the case of SBHSE, dissertations are typically organized with a general introduction, followed by studies written in the form of peer-reviewed publications, ending with a general conclusion (and any Appendices).

Format Approval

For format, Graduate College must review the final copy of the dissertation. Copies of the Format Manual are available at https://graduate.asu.edu/completing-your-degree/format. The student is required to submit appropriate signed forms and a complete copy of the thesis for format review at least 10 working days (two weeks if there are no holidays during the time period) before the oral defense.

The student must submit two final copies of the thesis. The student is responsible for the binding fees. Binding services are available through Arizona Library Binding at 602- 253-1861.

The process for completing the necessary degree requirements and steps can be found at the Graduate College website: https://students.asu.edu/graduation-apply.

Oral Defense of the Dissertation

The dissertation research experience culminates in a final oral exam, commonly known as the "dissertation defense." The final oral examination in defense of the dissertation is mandatory and must be held on the campus of Arizona State University. The student will complete the online Doctoral Defense Scheduling Form once they have identified a date and time with their committee. This form enables our Advising Office to officially schedule the defense. (Students are encouraged to email sbhse@asu.edu for further assistance if they do not get a response within 48 hours of completing the form, excluding weekends).

A final public dissertation defense is required. The defense must be scheduled officially with the Graduate College. Defenses that are held without being scheduled with the Graduate College are considered invalid. At least 50% of the committee must be physically present at the oral defense. Students must be physically present at the oral defense of their dissertation. A virtual defense option is only available to students in approved online programs. During the semester that the students defend the dissertation they are required to register for:

- At least one semester hour of credit that appears on the Interactive Plan of Study; OR
- At least one semester hour of appropriate graduate-level credit, for example: Research (792), Dissertation (799), or Continuing Registration (795); OR

For additional deadlines and dissertation requirements, visit https://graduate.asu.edu/current-students/policies-forms-and-deadlines/graduation-deadlines

At the beginning of the examination, the student's research advisor introduces the student and the topic of their research to the general audience. The student is then expected to present a brief seminar outlining the results of their research. The presentation should be limited to 45 minutes. Following the presentation by the student, the general audience is invited to ask questions. Following this question and answer session, the general audience is excused and the student's dissertation committee continues to question the student in depth regarding his/her research findings. The student should be prepared to defend the research methodology used in the study and the results obtained.

The oral defense of the dissertation is limited to a period of three hours. If necessary, however, the proceedings may be adjourned and rescheduled for a mutually convenient date within one week. Only one adjournment is permissible. When the dissertation committee completes its questioning, the student is asked to leave the room and the committee discusses whether or not the student successfully defended their research and whether or not the completed dissertation is acceptable.

Level of Pass or Fail

Pass: Only minor format corrections need to be made (e.g., typographical errors, and pagination). At the conclusion of the defense, 1) the committee chair should indicate "pass" and briefly describe needed revisions, and 2) all committee members should report the examination results at the bottom of form and sign the dissertation approval page.

Pass with minor revisions: Extensive format/editorial corrections and/or minor substantive changes need to be made (e.g., rewrite some text, correct grammatical errors). At the conclusion of the defense, 1) the committee chair should indicate "pass with minor revisions" and briefly describe revisions, and 2) the committee members, not including the chairperson, should report the examination results at the bottom of the form and sign the thesis approval page. 3) After revisions are made, the chairperson should report the exam results at the bottom of the form and sign the dissertation approval page.

Pass with major revisions: Extensive substantive changes need to be made (e.g., chapter rewrite). 1) At the conclusion of the defense, the committee chair should indicate "pass with major revisions" and briefly describe revisions. 2) After revisions are made, all committee members should report the examination results at the bottom of the form, and sign the dissertation page.

Fail: The basic design and/or overall execution of the study are flawed or the candidate's performance in the oral examination is seriously deficient, and beyond remediation as determined by the committee. This outcome is extremely rare. In this case, at the conclusion of the defense, 1) the committee chairperson should indicate "fail", and 2) all committee members should report the examination results at the bottom of the form. The dissertation approval page should not be signed. While this is very rare, if a student fails their dissertation defense, per the Graduate College recommendation, the student will be dismissed from the program. If the student is dismissed from the program, they may be eligible to apply for a Masters in Passing (MIP), pending eligibility.

The results of the oral defense are conveyed to the student by the chair of the supervisory committee or dissertation committee, whichever is appropriate. The results are transmitted to the Graduate College via submission through the iPOS. A guide for how to submit defense results in the iPOS can be found here. The Committee Chair and/or Co-Chair (ASU faculty only) have the option to enter defense results on behalf of a committee member or affiliate co-chair.

Revisions to the dissertation are typical and must be completed in a timely manner. If students are unable to complete revisions to the document and submit to UMI/ProQuest by the deadline for the semester in which the defense is held, they must complete the revisions, remain continuously enrolled and present the final document to UMI/ProQuest within one year of the defense. Failure to do so will require the re-submission of the document for format review and may result in re-defense of the dissertation to ensure currency of the research.

Applying for Graduation

The student is eligible for graduation when the final oral examination is passed and the dissertation is approved by the supervisory committee and accepted by the Director of the School of Biological and Health Systems Engineering and the Dean of the Graduate College.

The application for graduation should be submitted no later than the date specified in the Graduate College calendar (Refer to Graduate College website for current information). All fees are payable at this time. The student applies for graduation on their My ASU graduation tab.

An additional late fee will be assessed if the student applies after the deadline. If a student does not complete all degree requirements by the date of graduation for which they have applied, the graduation may be moved to a future term. The student should contact the graduation office to make this arrangement.

For information on University defense and graduation deadlines, visit: https://graduate.asu.edu/completing-your-degree

Enrollment

Students must be enrolled for at least one hour of credit that appears on the plan of study or one hour of appropriate graduate-level credit during the semester or summer session in which they defend a dissertation.

Summer: During the summer session, enrollment in any one of the summer sessions will fulfill the requirement.

Break Period. Students with an oral defense scheduled during a break period must be enrolled in both the proceeding semester and the following semester, including summer term. If the break is between the summer and fall, enrollment during any one of the summer sessions will fulfill the requirement.

Continuous Enrollment in a Doctoral Degree Program

Once admitted to a doctoral degree program, the student is expected to be enrolled continuously, excluding summer sessions, until all requirements for the degree have been fulfilled. Students must be enrolled in courses that meet the program requirements, which may include coursework, 792 Research, or 799 Dissertation. Credits that do not meet program requirements will not count toward continuous enrollment. If no additional credit is required toward the doctoral degree, the student typically continues to enroll in at least one credit of 792 Research until the dissertation defense is passed.

If a plan of study must be interrupted, the student may apply for leave status. The approved petition must be filed no later than the last day to register for classes in the semester for which the student is requesting a leave.

A student who interrupts a program without obtaining leave status will be removed automatically from the program.

International Students in the Final Semester

F1/J1 visa students who have less than 9 credits remaining in their final semester should submit a reduced course load e-form available at https://issc.asu.edu/f-1j-1-students/reduced-load.

Requesting a Leave of Absence

Graduate students planning to discontinue registration for a semester or more must submit a leave of absence request via their Interactive Plan of Study (iPOS). This request must be submitted and approved before the anticipated semester of non-registration. Students may request a maximum of two semesters of leave during their entire program. Students with a Graduate College-approved leave of absence are not required to pay tuition or fees, but in turn are not permitted to place any demands on university faculty or use any university resources. These resources include university libraries, laboratories, recreation facilities or faculty and staff time. For more information, please contact sbhse@asu.edu.

Medical/Compassionate Withdrawal

The medical and compassionate withdrawal process is focused on the student's academic record as it relates to the student's health and wellness. Tuition refunds are not guaranteed, even with approval. A medical/compassionate withdrawal request may be made in extraordinary cases in which serious illness or injury (medical) or another significant personal situation (compassionate) prevents a student from continuing his or her classes, and incompletes or other arrangements with the instructors are not possible.

If you experienced other challenges during the semester, such as difficulty with classes, time management, work or family responsibilities, or other co-curricular commitments, be aware that these are not considered extenuating circumstances. In these cases, consult your academic advisor and utilize ASU resources to ensure that you receive the guidance and assistance necessary to remain on track to graduate.

Medical and compassionate withdrawal requests are reserved for extraordinary and emergency circumstances that prevent a student from completing their classes. All requests require specific and relevant professional documentation for consideration, and approval is not guaranteed. The decision is based on the specific circumstances and the professional documentation provided. Approval is made on a case-by-case basis and is made at the discretion of the college. *The decision of the college is final.* More information can be found here on the College website.

Other Dissertation and Student Requirements upon Graduation

Doctoral students must complete all program requirements within a ten-year period. The ten-year period starts with the initial enrollment into the doctoral program. Any exception must be approved by the supervisory committee and the dean of the Graduate College and ordinarily involves repetition of the comprehensive examinations.

Other dissertation requirements. The student must submit three copies of the dissertation for binding. Bound copies are placed in the University Library, department office, and Archives. Bound copies of the dissertation are also prepared for the student's research advisor. Doctoral candidates must also submit one copy of the title page and one copy of the abstract (which must not exceed 350 words) to the bookstore.

Binding services are available through Arizona Library Binding at 602-253-1861. The student is responsible for the binding fees. Doctoral students must also pay to have their dissertations microfilmed by University Microfilms International (UMI).

Other requirements. Keys must be returned, all departmental property must be returned, samples and notebooks must be turned over to the advisor, wastes must be disposed of and the student's desk must be cleaned out.

Financial Support

Financial support for graduate students in the School of Biological and Health Systems Engineering is available from several sources. These include Graduate Research Assistantships (GRA), teaching assistantships (TA), and academic scholarships. In all cases, program fees are typically covered by the student themselves, which vary based on a number of factors, such as residence status.

Graduate Research Assistantships

Graduate Research assistantship (GRA) appointments typically pay the student a stipend to participate in a particular research project that may serve as his/her thesis research topic. The research assistant may be appointed 50% time (20 hours per week) or 25% time (10 hours per week). Students receiving stipends for research activity that also constitutes the dissertation research spend considerably more time each week working on the project than that dictated by the assistantship. The vast majority of PhD students in our program are supported through faculty-sponsored GRAs from external funding sources. Health insurance and tuition are covered with a GRA appointment.

Teaching Assistantships

Some teaching assistantships (TAs) may be available to qualified individuals on a semester-by-semester basis, although the majority of the PhD students in our program are supported through GRAs. Selected students receiving teaching assistantships may be assigned appointments that are half-time (20 hours per week) or quarter-time (10 hours per week). Assignments may include sole responsibility for the teaching of undergraduate laboratories, assistance in the teaching of undergraduate laboratories or assistance in the grading of undergraduate homework. Occasionally the student may be asked to prepare specific lectures in undergraduate courses and administer examinations. Teaching responsibilities are in addition to the time spent on research for the graduate degree. A tuition waiver is usually given to students awarded graduate assistantships. In some cases, a doctoral student may be listed as instructor of record for a course, and deliver the entire course. In addition to the above requirements, a student must also complete their Teaching Practicum prior to being listed as instructor of record. A student may complete their Teaching Practicum in the same semester in which they are receiving a teaching assistantship.

At this time, students selected for teaching assistantships are nominated by their faculty advisors through an internal form managed by the Graduate Program Chair. The final selection process is overseen by SBHSE leadership, prioritizing 1) BME PhD students, 2) students whose advisors have a primary appointment in SBHSE, 3) students who have been supported for at least two years already by their advisor or an external fellowship, and 4) students who have passed both the written and oral portions of their Comprehensive Exam.

Scholarships

The Graduate College provides a variety of mechanisms to support funding for outstanding graduate students recommended by the program if funds are available (see http://graduate.asu.edu/financing). Students may apply for these awards provided by the Graduate College. Generally students receiving research assistantships or teaching assistantships qualify for out-of-state tuition waivers. Only a very limited amount of support is available. These are awarded to the students with the most outstanding academic credentials. The Fulton Schools of Engineering also provides scholarship opportunities, for which the applications typically close mid-Spring.

Policies Related to Financial Support of Graduate Students

It is the desire of the School to provide financial support for as many students as possible, most commonly through GRAs. Thus, a limited number of students receive written offers of financial aid in the form of a GRA prior to entering the program. Eligible applicants are admitted into the PhD program under the supervision and support of a faculty advisor with a primary appointment in the School of Biological and Health Systems Engineering and with membership to its graduate faculty. Eligible applicants can be admitted under the supervision and support of a faculty member with a primary appointment outside of the School, either in another academic unit within ASU or at an established clinical partner, who agree to the processes and procedures outlined in this Handbook. In all cases, financial support is provided through the faculty advisor in the form of a Graduate Research Assistantship (GRA) during the academic year, with the advisor's funding confirmed prior to admission. Applicants without funding confirmation and commitment of funding from a faculty advisor ***or*** an external academic fellowship (e.g., NSF GRFP, GEM Fellowship, which must be confirmed in writing) will not be admitted. Any eligible applicant who chooses to maintain additional employment in industry must finalize all necessary legal agreements between their employer and ASU prior to being admitted into the program; often these agreements take one or more years to be completed. It is understood that any suspension from the graduate program results in the loss of financial support. Finally, departmental decisions on financial aid over the course of their degree program once admitted are based on consideration of all aspects of each individual student's situation within the framework of these guidelines.

Intellectual Property

Key intellectual property policies can be found within the Arizona Board of Regents Policy Manual as well as ASU's Research and Sponsored Projects Manual. It is both the student's and faculty advisor's responsibilities to understand and remain in compliance with these key policies. These policies confirm and clarify ownership of research data and materials. For additional information, visit https://www.asu.edu/aad/manuals/rsp/rsp604.html

Conflict of Interest

In some cases, students can find themselves working on projects which are part of a commercial development, either of their own, or associated with a faculty member. Once a conflict of interest has been identified, the student and faculty member must complete the necessary COI steps in the MyDisclosures portal in the Enterprise Research Administration (ERA) system. Individuals should contact **MyDisclosures@asu.edu** with any further questions. Please note that University Counsel may also be involved, depending on the nature of the commercial development.

Access to SBHSE Staff and Facilities

ISAAC and Building Access

ISAAC (key card) provides access for the offices and laboratories in the Ira A. Fulton Schools of Engineering: Engineering Research Center (ERC), ISTB1, ISTB4, PEBE, Schwada (SCOB) Classroom Office Building, and Goldwater Center (GWC) are obtained by completing an online application, available at https://fultonapps.asu.edu/isaac/.

Please note that you must be either located on campus or logged in via an ASU recognized VPN program to access this website. The student's research advisor and an authorized department signor must approve the online form. ISAAC access will be granted to your ASU Sun Devil ID Card.

Office Equipment

Graduate students are not permitted to use office resources (computers, printers) without departmental approval. Students are urged to familiarize themselves with the extensive free computer facilities on campus available for word processing.

Copier and other Office Resources

The SBHSE copier is for faculty and staff use. Faculty may authorize their students to use the copier for teaching duties or for research. Large jobs (greater than 100 copies) require approval by the Business Operations Manager. No personal copying can be done on the SBHSE machine. Pay copiers are available at many locations on and off campus.

Misuse of departmental telephones, copiers, supplies, facilities is a serious offense that will lead to disciplinary action. At a minimum, students found to have used departmental resources for non- department approved purposes will be required to reimburse the department for such uses.

Additional University and Student Support Resources

FSE Academic Program Support

Graduate students in the SBHSE have access to the Fulton Schools of Engineering Graduate Programs website, which houses college resources and advising information. Graduate students are also encouraged to explore student organizations through the Graduate and Professional Student Association (GPSA). A one-page guide to Financial, Social, Emotional, and Physical Health and Wellness Resources for ASU Graduate Students, developed by the GPSA.

Student grade grievance appeals must be processed, by commencement, in the regular semester immediately following the issuance of the grade in dispute (fall or spring commencements only), regardless of whether the student is enrolled at the university. This process does not address academic integrity allegations, faculty misconduct or discrimination. The Fulton Schools of Engineering grade appeal procedures are based on the universities policy that can be found here. Students must begin with and complete the informal process prior to any decision on whether a formal hearing is warranted.

University Resources

- Graduate College
- Office of the University Provost

University-Wide Academic and Career Support

- ASU Libraries, including Noble Engineering Library
- Graduate Writing Center
- Career and Professional Development Services
- Graduate and Professional Student Association
- Fulton Schools of Engineering Student Clubs and Organizations

Business and Finance Services

- University Financial Aid and Scholarship Services (financial aid)
- Student Business Services (tuition, fees, and payments)
- Parking and Transit Services (permits, shuttles, public transit)
- Sun Devil Card Services (ID cards)
- University Technology Office (technology assistance)
- Sun Devil Dining (meal plans, M&G, hours)

Counseling Services

ASU Counseling Services provides confidential, time-limited counseling and crisis services for students experiencing emotional concerns or other factors that affect their ability to achieve their goals. Support is available 24/7.

In-person counseling

Monday-Friday 8 a.m. – 5 p.m.

ASU Counseling Services, Student Services Building 234 Tempe, AZ 85287

480-965-6146

After-hours/weekends

Call EMPACT's 24-hour ASU-dedicated crisis hotline: 480-921-1006

For life threatening emergencies: Call 911

The Grad College has also compiled a one-page quick sheet on 10 Best Practices in Graduate Student Wellbeing.

Disability Accommodations

Reasonable accommodations are determined on a case-by-case, course-by-course basis to mitigate barriers experienced due to a disability (SSM 701-02). Students with disabilities who require accommodations must register with the Student Accessibility and Inclusive Learning Services and submit appropriate documentation. It is recommended students complete this process at the beginning of the term and communicate as appropriate with their instructor.

- Email: Student.Accessibility@asu.edu
- Phone: (480) 965-1234
- FAX: (480) 965-0441

Pregnancy: Students requesting services due to pregnancy (SSM 701-10) should be prepared to submit documentation regarding the pregnancy, any complications and clearance to return to school related activities. Student Accessibility can work with students to foster continued participation in a program, whether that be with academic accommodations such as absences or assistance requesting a leave, or through other requested accommodations.

Health and Fitness

All ASU students enrolled in in-person programs have access to Sun Devil Fitness facilities on all campuses. For more information about facilities, membership and group fitness classes, please visit: https://fitness.asu.edu

For information about health insurance and appointments with care providers, please see the ASU Health Services website: https://eoss.asu.edu/health

International Students

ASU's International Student and Scholars Center can provide support and answers to questions about visas, employment, scholarships and travel. To find more information or schedule an appointment with an ISSC adviser, visit the website: https://issc.asu.edu/

Veterans and Military

The Pat Tillman Veterans Center provides guidance and support for students who are veterans, active-duty military or military dependents. For more information, please call the office at 602 496-0152 or visit: https://veterans.asu.edu/